Cefazolin

Mechanism of action

- Cefazolin is a first-generation cephalosporin antibiotic with activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Sepsis, where the source of infection appears to be the soft tissues or joint, and:
  - The patient is aged greater than or equal to 12 years, and
  - Has one or more clinical features indicating antibiotics, and
  - Time to hospital is greater than 30 minutes.
- ✓ Compound fracture
- ✓ Large contaminated wound.
- ✓ Following chest decompression.
- ✓ Following amputation.
- ✓ Cellulitis. In this setting a single dose may be administered if the patient is not being directly referred (or transported) to a medical facility.

Contraindications

- ✗ Known severe allergy to cephalosporins.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

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Dosage

- 1 g IV for an adult.
- See the paediatric drug dose tables for a child.

Administration

- Add approximately 4 ml of 0.9% sodium chloride to a 1 g ampoule and shake until dissolved.
- Draw up the ampoule contents and dilute to a total of 10 ml.
- Administer IV over 1-2 minutes preferably into a running line.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes

Usual duration of effect

- 6-8 hours.

Usual preparation

- Ampoule containing 1 g of cefazolin as a powder for reconstitution.

Pharmacokinetics

- Cefazolin is predominately excreted in the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.

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Common interactions

- None.

Additional information

- Rapid administration of cefazolin can cause pain and phlebitis, which is why it should be preferably administered into a running line.

More Information

Version history

28 October 2024 1.0.5
Update: Indications updated.

11 September 2023 1.0.4
New: Guideline released.